How can a function be represented

- **On a grid?** Favoured by scientist working with Numerical methods
- + Easy and intuitive. Banded matrices. Finite expansion.
- - Expensive. Not very accurate: numerical derivation and integration. Limited flexibility (a different grid gives a different equation).
- **Global analytical function?**. As Harmonic Oscillator functions to describe something approximately an Harmonic oscillator. Or Gaussians. Traditional method used in Physics.
- + Well chosen! $\rightarrow$ minimal expansion! Analytical integration and derivation. Accurate and potentially fast.
- - Infinite expansion - where to stop? Usually not a banded matrix.

$$\Phi(x) = \sum_{n=1}^{\infty} c_n \psi_n(x)$$

How can a function be represented

- **With localized analytical functions?** Examples: Finite Elements, **B-splines**.
- + Accuracy: Analytical integration and derivation. Adaptable to changing conditions. Finite expansion. Banded matrices.
- - Not as intuitive as e.g. a grid.

B-splines: piecewise polynomial functions

$$B_{i,k=1} = 1 \text{ if } t_i \leq x < t_{i+1}, \text{ otherwise } B_{i,k=1} = 0$$

$$B_{i,k}(x) = \frac{x - t_i}{t_{i+k-1} - t_i} B_{i,k-1}(x) + \frac{t_{i+k} - x}{t_{i+k} - t_{i+1}} B_{i+1,k-1}(x),$$

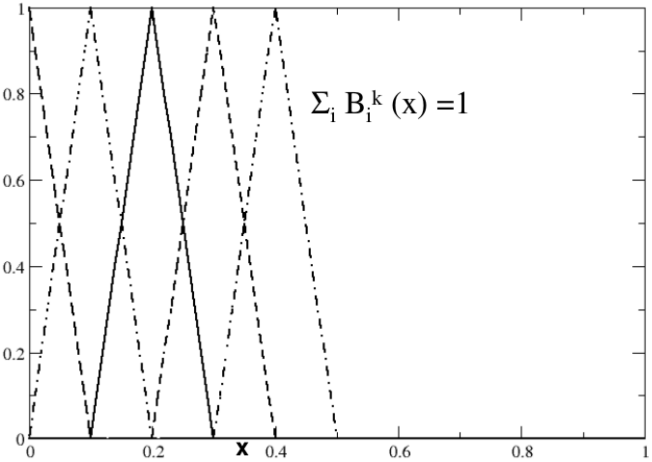
where the knot-point numbering *includes* any extra *ghost* points placed in the first and last points. **No restriction on the "grid"!**

FIGURE: With N knot points (here 17), and $2(k-1)$ ghost-points, we get $(N-2k+2)$ physical points (here 11) and $(N-k)$ B-splines (here 13). With k knots in first and last point all B-splines are zero outside the considered region

• Convention: if the numerator is zero the B-spline is zero, even if the denominator is also zero

B-splines with k=2

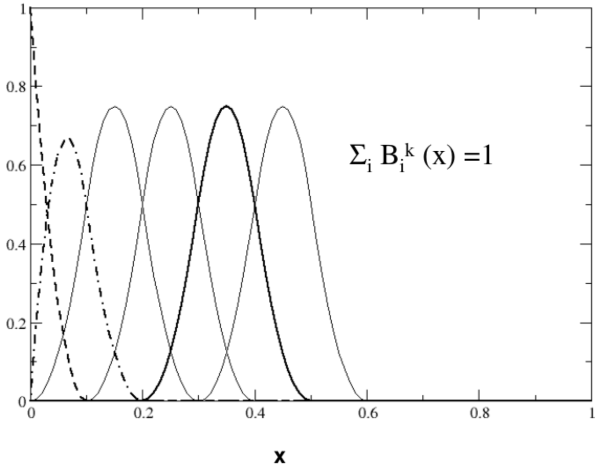
Continuous splines, discontinuous derivatives



knots in 0.0, 0.1, 0.2, 0.3 etc., k=2 knots in first and last point

B-splines with k=3

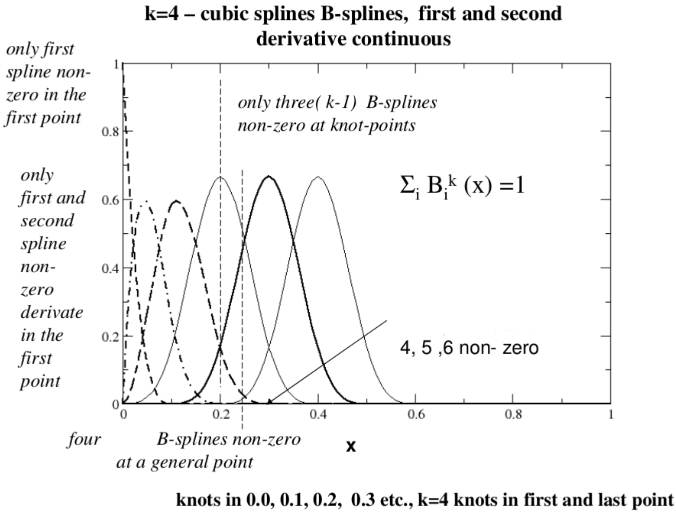
Continuous splines, continuous derivatives, discontinues second order derivatives



knots in 0.0, 0.1, 0.2, 0.3 etc., k=3 knots in first and last point

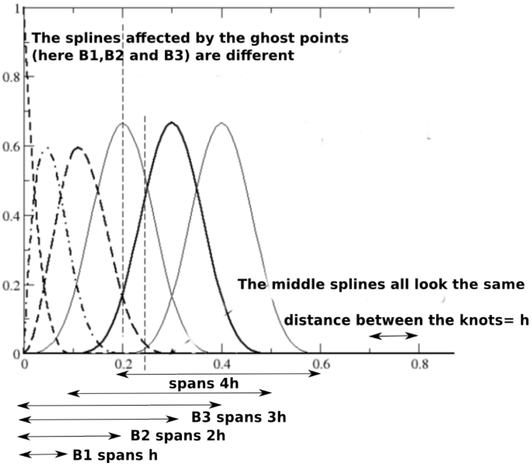
B-splines with k=4

Continuous splines, continuous first and second order derivatives, discontinues third order derivatives



The knot sequence

- B-spline i (B_i^k) starts in knot point i (t_i) and ends in t_{i+k}
- To confine the B-splines to a limited space you need to put several knot points in the same physical point. This applies to both the starting of the knot and the ending. Examples below for the starting only.
- Typically with $k = 4$ and spherical coordinates you have 4 knot points in the origin. $t_1 = t_2 = t_3 = t_4 = 0$. The first three can be called *ghostpoints*. If you chose a linear grid the remaining points are at $t_5 = h$, $t_6 = 2h$ etc.
- with $k = 4$ B_1 starts in t_1 and ends in t_5, B_2 goes from t_2 to t_6, B_3 goes from t_3 to t_7, and B_4 goes from t_4 to t_8. While B_4,B_5 etc all span a distance of $4h$ the first three are different. B_1 span the distance h, B_2 $2h$, and B_3 $3h$. These three splines look thus *different* from the rest. This is necessary for the completeness relation to hold: $\sum_i B_i(x) = 1$.



The knot sequence determines the B-splines. With 4 knot points in the first physical point the boundary condition that $f(x) = \sum c_i B_i(x)$ is zero in $x = 0$ is implemented by putting $c_1 = 0$. With other knot sequences it has to be differently implemented.

Collocation

The B-splines form a basis in which we may expand the unknown function f : $f(x) = \sum_i c_i B_i^k(x)$. The B-splines are only non-zero over k knot-intervals. Calling the first knot point #1 (the first physical point is thus #k), then if $t_i \leq x \leq t_{i+1}$, then a simpler relation holds:

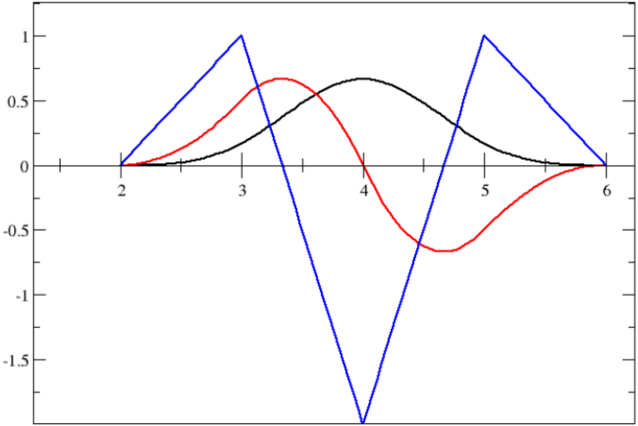
$$f(x) = \sum_{n=i-k+1}^{i} c_n B_n^k(x), \text{ for } t_i \leq x \leq t_{i+1}$$

which at the knot points is even simpler:

$$f(t_i) = \sum_{n=i-k+1}^{i-1} c_n B_n^k(t_i)$$

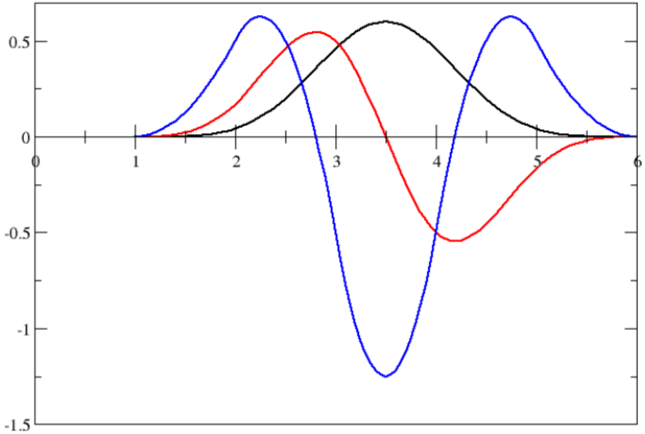
To find the c:s. Use your equation and the boundary conditions to set up a system of $N - k$ linear equations with the c:s as the unknowns. Solve the system of linear equations with standard methods.

B-splines with k=4



Bspline # 6, k=4, its derivative, and second order derivative

B-splines with k=5



Bspline # 6, k=5, its derivative, and second order derivative

Collocation - solve a boundary value problem

$$\frac{\partial^2 f(x)}{\partial x^2} + p(x)\frac{\partial f(x)}{\partial x} + q(x)f(x) = g(x), \;\; a \leq x \leq b$$

with boundary conditions

$$f(a) = \alpha, \;\; f(b) = \beta.$$

$$f(x) = \sum_n c_n B_{n,k}(x).$$

- second order differential equation, $k \geq 4 \rightarrow$
- B-splines twice differentiable
- function, derivative, 2nd order derivative continues.
- Require the equation to be fulfilled in specific points: **collocation!**

Collocation - solve a boundary value problem

- Equation to be fulfilled in specific points: **collocation!**
- use the knot points (nothing special about them, but it gives minimum non-zero elements and non-zero rank of the matrix)
- N knots points, k knots points in first and last physical point
- $\rightarrow N - 2 \times (k-1)$ physical points, $(N-k)$ B-splines.
- one Eq. per knot point: $N - 2 \times (k-1)$ equations
- $k-1$ B-splines non-zero at the knot points, each equation involves $(k-1)$ B-splines. $k = 4 \rightarrow$ **tridiagonal matrix**.

$$f(x_i) = \sum_{n=i-k+1}^{i-1} c_n B_{n,k}(x_i),$$

- $k = 4 \rightarrow (N-6)$ equations and $(N-4)$ unknown c_n
- use the 2 boundary conditions $\rightarrow (N-4)$ equations

A MATRIX EQUATION

- A system of $N-4$ linear equations!

$$\mathbb{A}\mathbb{C} = \mathbb{B}$$

- $\mathbb{A}$ banded matrix
- $\mathbb{B}$ the right-hand side
- $\mathbb{C}$ column vector with the coefficients c_i.

The Poisson equation with spherical symmetry

$$\nabla^2 V(r) = -\frac{4\pi\rho(r)}{4\pi\varepsilon_0}, \qquad \nabla^2 V(r) = \frac{\partial^2 V(r)}{\partial r^2} + \frac{2}{r}\frac{\partial V(r)}{\partial r}$$

Choose $V(r) = \frac{\varphi(r)}{r}$

$$\rightarrow \nabla^2 V(r) = \frac{1}{r}\frac{\partial^2 \varphi(r)}{\partial r^2} = -\frac{4\pi\rho(r)}{4\pi\varepsilon_0},$$

$$\frac{\partial^2 \varphi(r)}{\partial r^2} = -r\frac{4\pi\rho(r)}{4\pi\varepsilon_0},$$

The ansatz is now

$$\varphi(r) = \sum_n c_n B_n^k(r) \quad \rightarrow \qquad \sum_n c_n \frac{\partial^2 B_n^k(r)}{\partial r^2} = -r\frac{4\pi\rho(r)}{4\pi\varepsilon_0}.$$

- fulfilled at the knot points

$$\sum_{n=i-3}^{i-1} c_n \left.\frac{\partial^2 B_n^{k=4}(r)}{\partial r^2}\right|_{r=r_i} = -r_i\frac{4\pi\rho(r_i)}{4\pi\varepsilon_0}.$$

Collocation: the matrix

Here we assume $k = 4$ and count the knot points from the first *ghost point*. The first physical knot point is thus $t_4 = 0$. We assume that the boundary condition is that the solution is zero at the origin and have removed B_1. B'' denotes the second order derivative with respect to r.

$$\begin{pmatrix} B_2''(t_4) & B_3''(t_4) & 0 & \dots & & \\ B_2''(t_5) & B_3''(t_5) & B_4''(t_5) & 0 & \dots & \\ 0 & B_3''(t_6) & B_4''(t_6) & B_5''(t_6) & 0 & \dots \\ 0 & 0 & B_4''(t_7) & B_5''(t_7) & B_6''(t_7) & \dots \\ \dots & & & & & \end{pmatrix} \begin{pmatrix} c_2 \\ c_3 \\ c_4 \\ c_5 \\ \dots \end{pmatrix} = \begin{pmatrix} rhs(t_4) \\ rhs(t_5) \\ rhs(t_6) \\ rhs(t_7) \\ \dots \end{pmatrix}$$

The unknowns to solve for are the coefficients c. The boundary condition at $r = 0$ is implemented $(c_1 = 0)$. To implement that the derivative of the solution is zero at the last knot add the equation:

$$c_{N-4} B'_{N-4}(t_{last-3}) + c_{N-5} B'_{N-5}(t_{last-3}) = 0$$

Note: Number of knotpoints N. t_{last-3} last is the last physical point. Number of Bsplines $N - 4$.

Solve for:

- Uniformly charged sphere
- Uniformly charged shell (with finite thickness)
- Hydrogen ground state charge distribution:

$$\frac{4\pi\rho(r)}{4\pi\varepsilon_0} = \frac{e}{4\pi\varepsilon_0}\, 4\pi\, \psi^*(r)\, \psi(r),$$

- Charge distribution with discontinuous derivative? How to choose the knot sequence?

What next?

- We will return to Poisson's equation in Assignment 7. If you make sure that you get the hydrogen ground state charge distribution correct, with proper normlization etc., you will already have one piece for that larger program in place.

Something to be aware of

Strictly from the definition the last B-spline

$$B_{last}(x) \to 1,\ x \to t_{last}$$

but it is zero exactly at the last physical point. This is impractical. The original routine by deBoor defines instead the last B-spline to be one even there. Then the condition that $\sum_i B_i(x) = 1$ is fulfilled for $t_{first} \leq x \leq t_{last}$. Routines that you find at other places might however not do that. Remember in that case that the last boundary condition can be implemented in $x = t_{last} - \delta$, with δ being a small number, instead of exactly in t_{last}.